

Online Experiments for Language Scientists

Lecture 4: Self-paced reading

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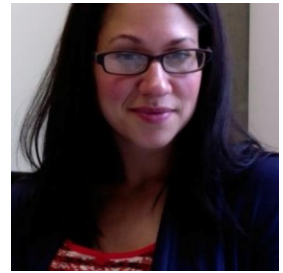
Based on the slides of Kenny Smith

Enochson & Culbertson (2015)

Enochson, K., & Culbertson, J. (2015). Collecting Psycholinguistic Response Time Data Using Amazon Mechanical Turk. *PLoS ONE*, 10, e0116946.

Three self-paced reading experiments on MTurk

- Do small but meaningful RT differences seen in lab studies replicate online, despite reduced experimental control and increased variability in e.g. participant hardware?

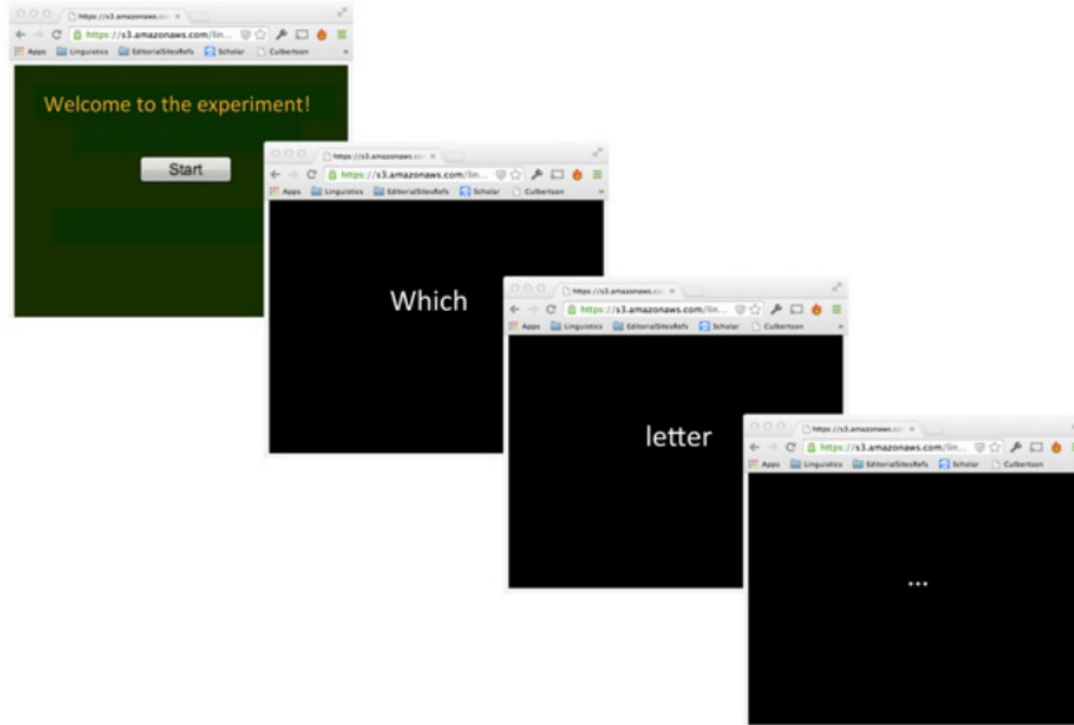


Jenny Culbertson
(Edinburgh)



Kelly Enochson
(formerly George Mason University)

Self-paced reading



Self-paced reading

Demo with this week's lab code

Sample size, study duration etc.

- Self-reported native speakers of English
- N=34 (Exp 1), 82 (Exp 2), 60 (Exp 3)
- 96-120 items per experiment (mainly fillers)
- 20 minutes, \$1

Test items and predicted effects (Exp 1)

Filler-gap (in all sentences)

- ***Which antique** was the maid polishing ____ in the study?*

slowdown at or
following a gap

Full DP vs pronoun

- *Which antique was **the maid** polishing in the study?*
- *Which antique was **she** polishing in the **upstairs** study?*

slowdown at the
subject

Agreement attraction

***Which antique** was the maid polishing in the study?*

***Which antiques** was the maid polishing in the study?*

slowdown somewhere
after the verb

Residual reading times

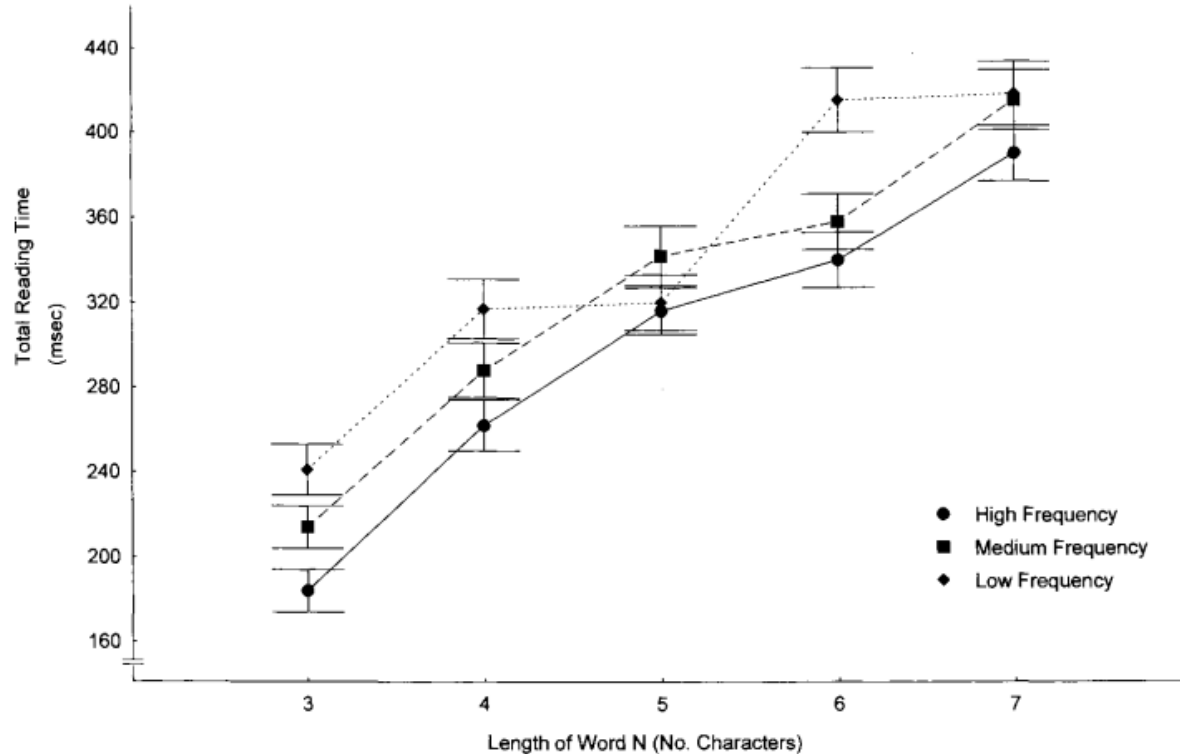


Fig. 1. Effects of length and frequency on word *N* reading time.

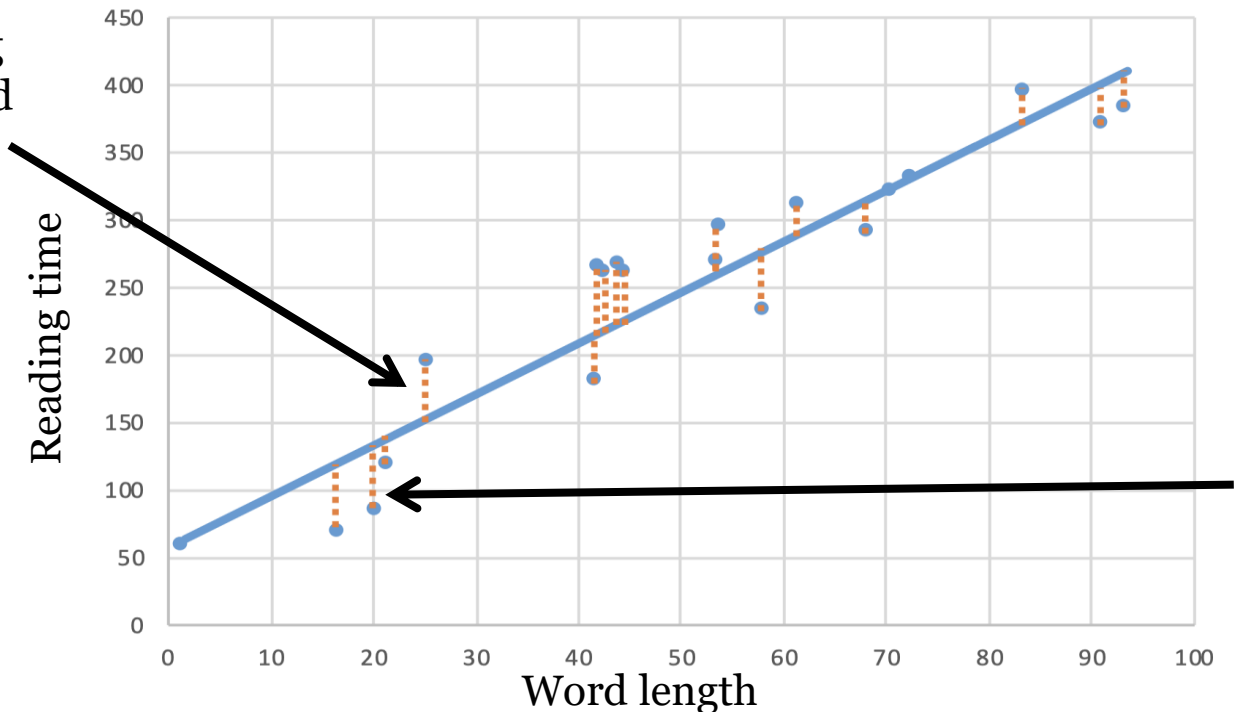
You would expect
e.g. word length and
frequency to
influence reading
time

Taken from: Underwood, G.,
Binns, A., & Walker, S. (2000).
Attentional Demands on the
Processing of Neighbouring
Words. In I. Kennedy, R.
Radach, D. Heller, & J. Pynte
(Eds.), *Reading as a
Perceptual Process* (pp 247-
268). Amsterdam: North-
Holland.

Residual reading times: illustration

Regression line – line of best fit, **minimising residuals**

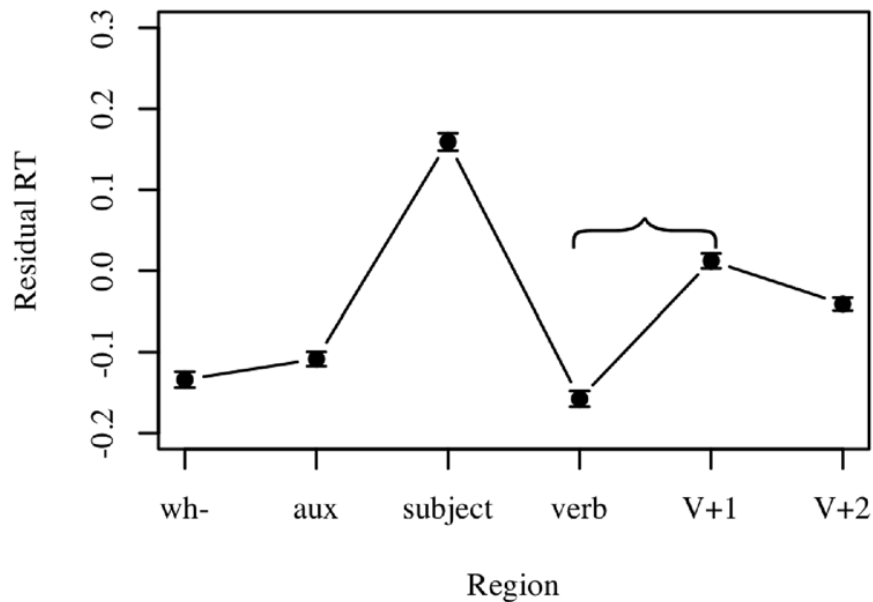
Slower reading
than you would
expect given
word length



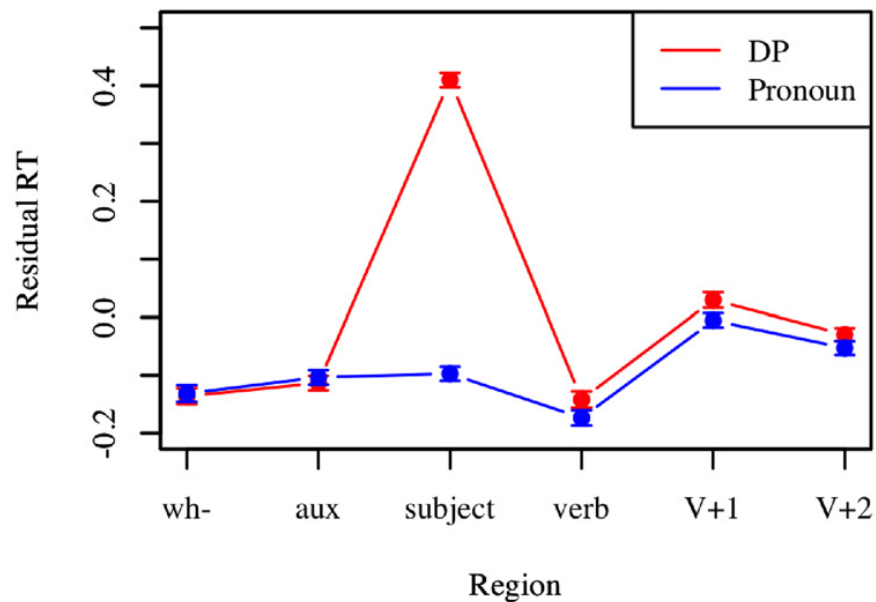
Faster reading
than you would
expect given
word length

Exp 1 results

Filler-gap effect



Pronoun vs. DP effect



Test items and predicted effects (Exps 2-3)

Experiment 2

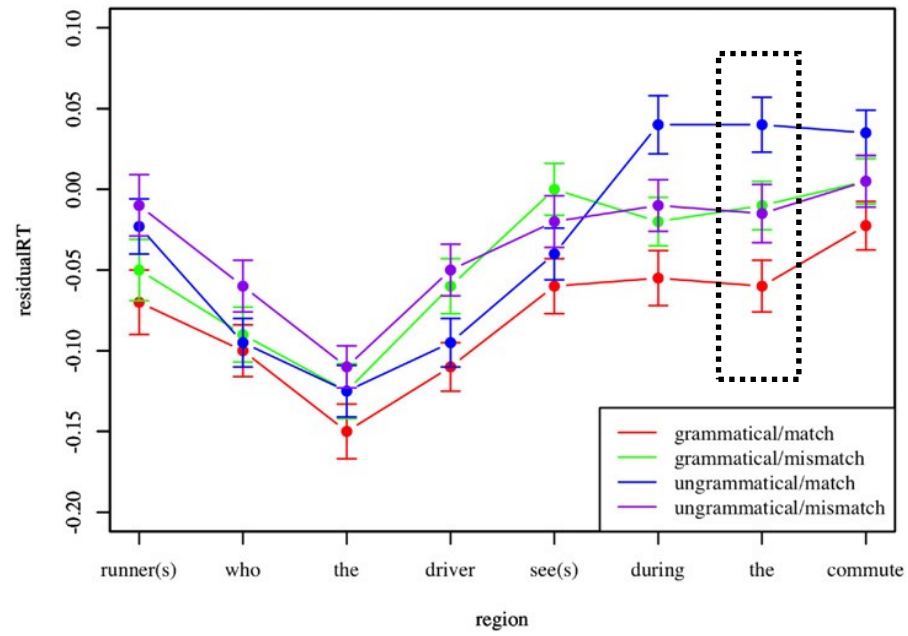
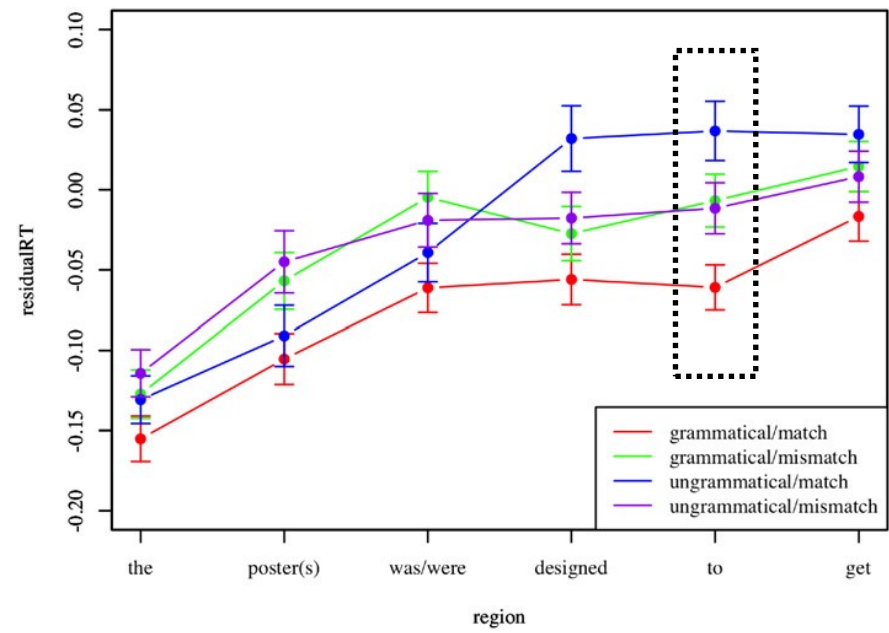
- *The slogan on the **poster** was designed to get attention* grammatical+match
 - *The slogan on the **posters** was designed to get attention* grammatical+mismatch
 - **The slogan on the **poster** were designed to get attention* ungrammatical+match
 - **The slogan on the **posters** were designed to get attention* ungrammatical+mismatch
- ↑ speed ↓

Experiment 3

- *The runner who the driver sees during the commute...* grammatical+match
 - *The runners who the driver sees during the commute...* grammatical+mismatch
 - ****The runner** who the driver see during the commute...* ungrammatical+match
 - ****The runners** who the driver see during the commute...* ungrammatical+mismatch
- ↑ speed ↓

The slogan on the poster was designed to get attention
The slogan on the posters was designed to get attention
**The slogan on the poster were designed to get attention*
**The slogan on the posters were designed to get attention*

The runner who the driver sees during the commute...
The runners who the driver sees during the commute...
**The runner who the driver see during the commute...*
**The runners who the driver see during the commute...*



Enochson & Culbertson's conclusions

MTurk is suitable for collecting reading-time data in self-paced reading tasks

- Similar patterns of results to those seen in lab tasks
- (Paper also includes lab replication of Exp 1)

They also make some suggestions re. Masters qualifications and batch sizes that not everyone necessarily agrees with – see the reading notes!

Time for Q&A/discussion on this week's
reading

Next up

Thursday lab

- Our second proper experiment: self-paced reading
- If you are behind, do your best to get caught up before the lab

Next week: Break!

- Good opportunity to make sure you're following what we have covered so far, before it gets more complicated!
- Week 6: Word learning/frequency learning